

Panda BOX Technical Specification

1 Description



Panda BOX is a 4G/WiFi/Bluetooth Industrial designed (ODM/OEM) for tank truck which use LCR/TCS meters monitoring. It is able to monitor vehicle location, fuel information from LCR/TCS meters with print function. It can act as an in-Vehicle WiFi hot spot, up to 8 WiFi enabled devices can be connected to share high speed 4G LTE network.

2 Features

- ◆ Industrial class components with high quality
- ◆ Compact size and easy installation
- ◆ High speed 4G LTE Cat. 4
- ◆ Bluetooth 4.2 BR/EDR and BLE5.0
- ◆ WiFi 2.4G data transmission with WiFi hot spot
- ◆ Wide operating voltage for heavy duty
- ◆ 32MB Memory FLASH, store 262,144 data packages
- ◆ Multiple interfaces for external meters monitoring and print control
- ◆ OTA FW upgrade

3 Functions

- ◆ Real-time tracking
- ◆ Read history record time
- ◆ Read BOX storage status
- ◆ Read BOX storage data by timestamp
- ◆ Delete storage data
- ◆ Bluetooth, WiFi, TCP server parameters configuration
- ◆ Read BOX status information
- ◆ Reset/start the BOX
- ◆ Set/read BOX time
- ◆ Start, Pause, Stop the meter
- ◆ Print control
- ◆ Set, read, modify meter address
- ◆ Pre-set gallon value
- ◆ Read the meter connection status
- ◆ Read the meter current working status
- ◆ Report data flow (gallon value), totalizer values (gallon value), GrossTotal_WM, NetTotal_WM, PreviousGross, PreviousNet in real time.

4 Application

- ◆ Tank truck monitoring (ODM/OEM)

5 Technical Specification

Mechanical	Size	158mm (L) x99mm (W) x 39mm (H)
	Weight	350g
	Communication Port	3x DB25, 1x DB9, 1x Micro USB
	Configuration Port	Micro-USB
	SIM Card Socket	Micro SIM, Push-Push type
Data Transmission		LTE/WCDMA, Bluetooth, WiFi, Serial
Positioning Method		GPS

Storage		32MB FLASH, store 262,144 data packages
Power	Working Voltage	9-36VDC
	Working Current	Max.: <200mA@13.8/27.6VDC Average: <150mA@13.8/27.6VDC Sleep mode: <5mA@12/24VDC
	Button Cell	3.7V/40mAH
GPS		Channels: 66 Sensitivity: -165dBm Accuracy: 5m CEP Time to first fix: Cold start: <30s (typ.) Hot start: <1s (typ.)
WiFi		Frequency: 2.4-2.4835GHz WLAN Standard: 802.11b/g/n Transmission Data: 65Mbps@802.11n 54Mbps@802.11g 11Mbps@802.11b Transmission distance: 120M
Bluetooth		Specification: Bluetooth low energy Frequency: 2.4-2.48GHz Modulation: GFSK (Bluetooth 4.2BR/EDR + 5.0BLE) Range: 10m (Max. no shield) Sensitivity: -93dBm@BLE Transmission distance: 100M
Cellular		Frequency (X-BOXLE, Nepal): FDD LTE B1/B3/B5/B7/B8/B20 TDD LTE B38/B40/B41 WCDMA/HSPA+ B1/B5/B8 GSM/GPRS 900/1800MHz Frequency (XBOXLA, US): FDD LTE B2/B4/B12 WCDMA/HSPA+ B2/B4/B5

		Data: LTE-FDD Max 150Mbps (DL)/50Mbps (UL) LTE-TDD Max 130Mbps (DL)/35Mbps (UL) HSPA+ Max 42Mbps (DL)/5.76Mbps (UL) Network protocol: Embedded TCP/IP stack
LED Indication		Cellular/WiFi/Bluetooth/GPS/Power/LCR Meter indication
Antenna	Cellular	External
	WiFi	Internal
	Bluetooth	Internal
	GPS	External
Environment	Working Temperature	-30°C ~ +80°C
	Storage Temperature	-40°C ~ +85°C
	Humidity	5% ~ 95% (no fog)